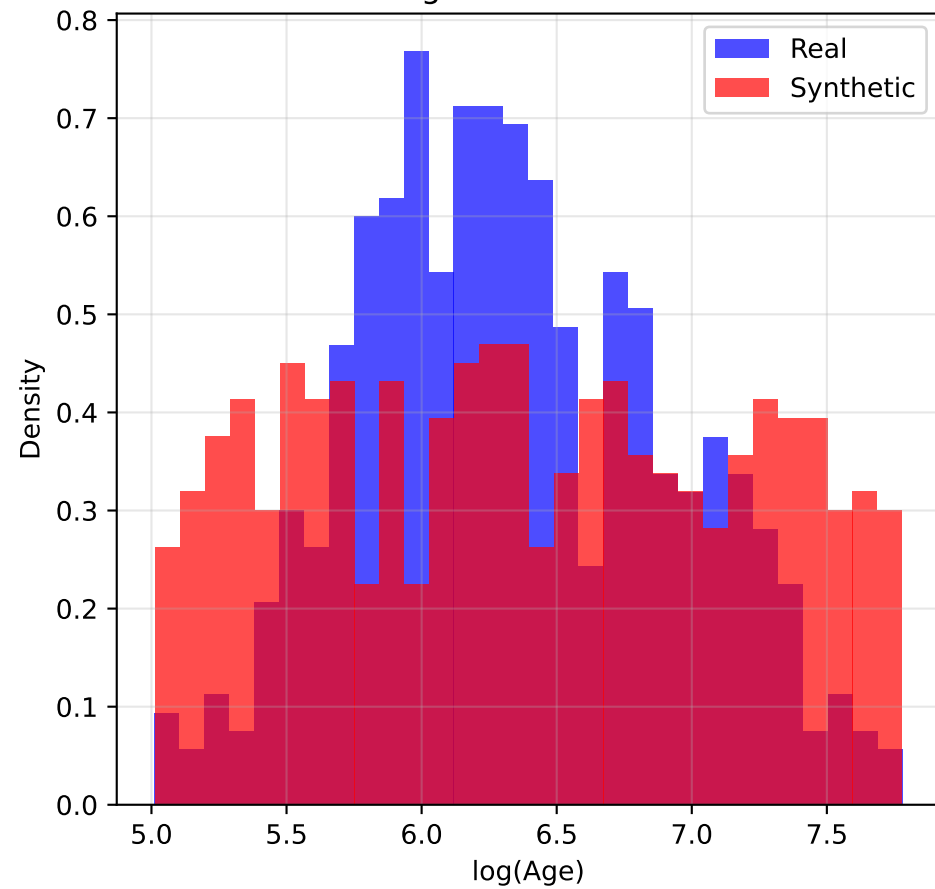
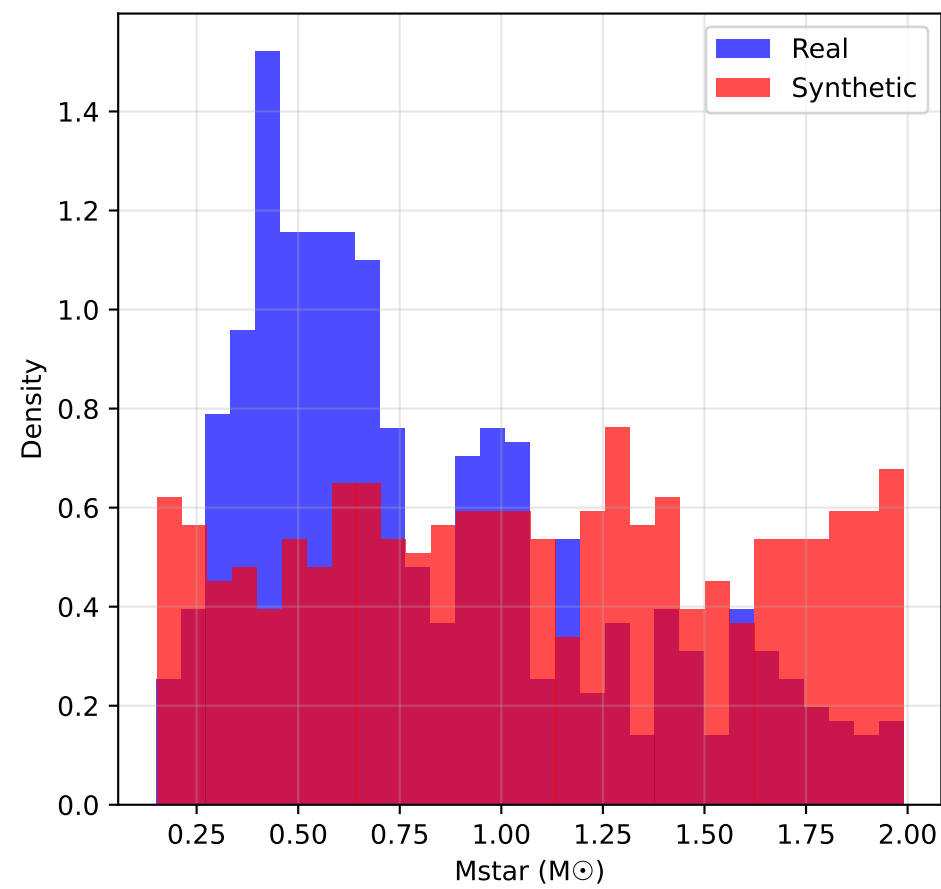


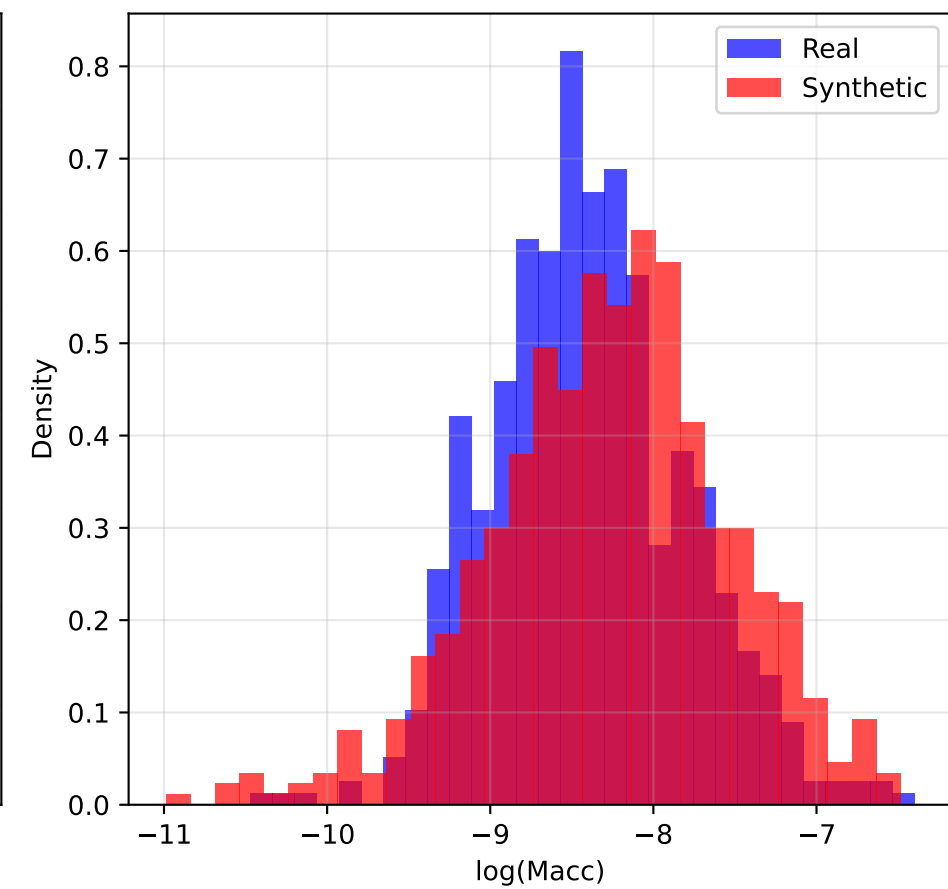
Age Distribution



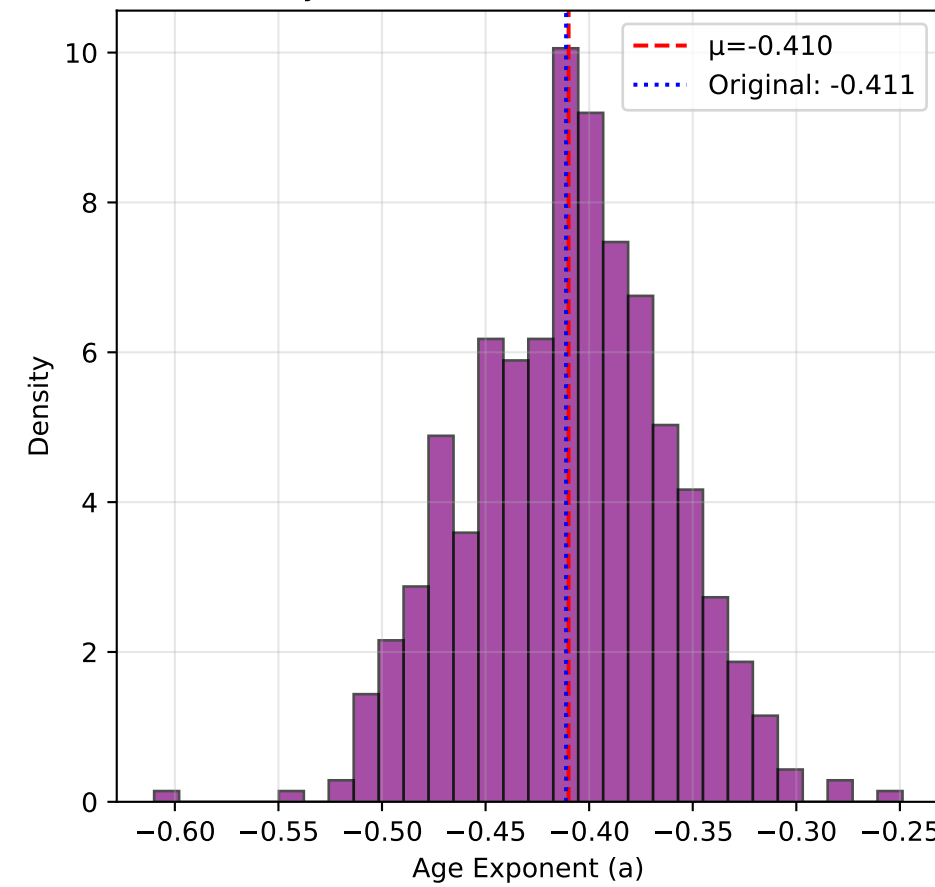
Mass Distribution



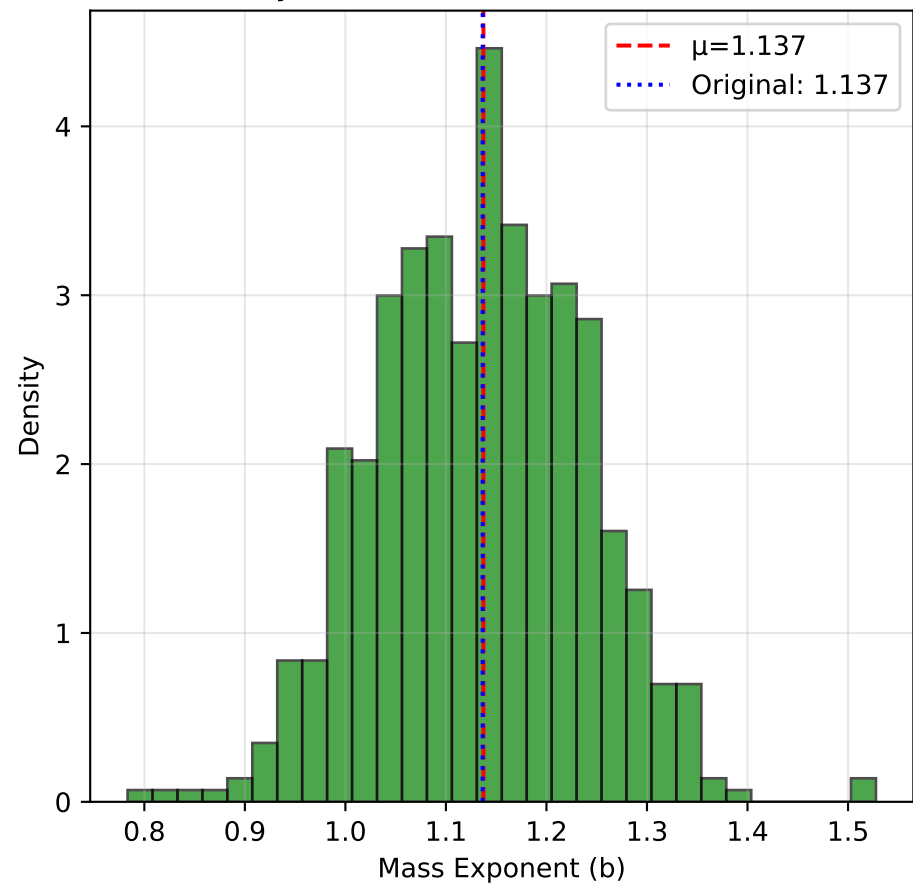
Accretion Rate Distribution



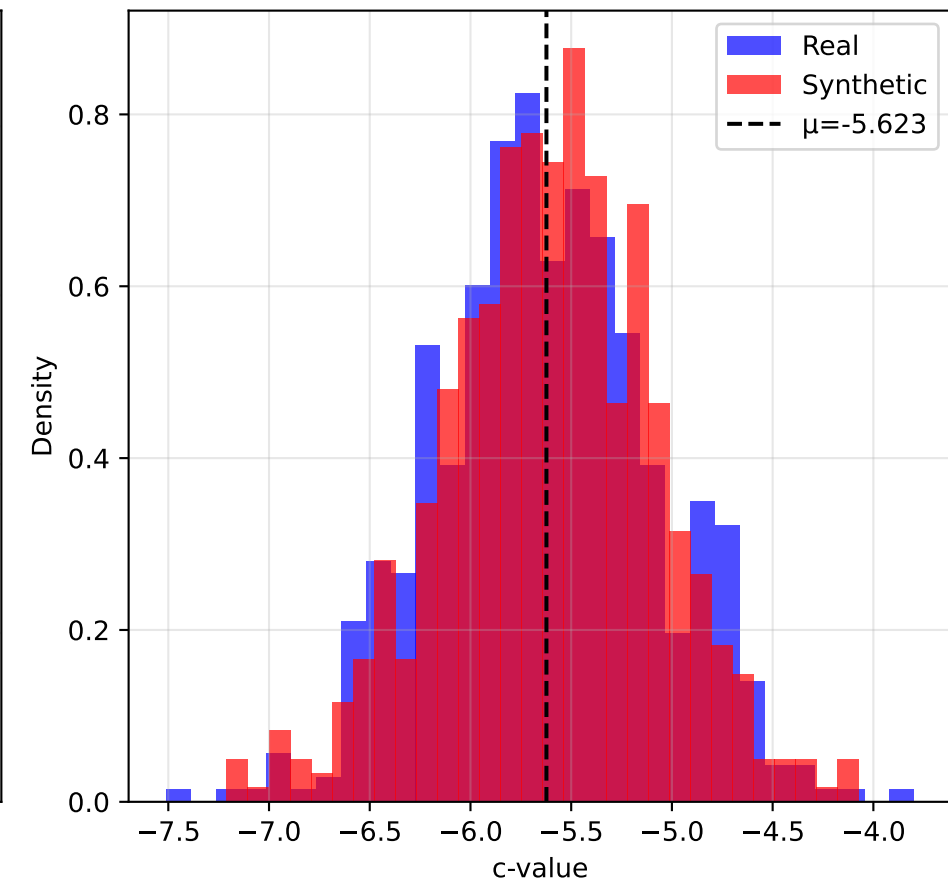
Synthetic a-values Distribution



Synthetic b-values Distribution

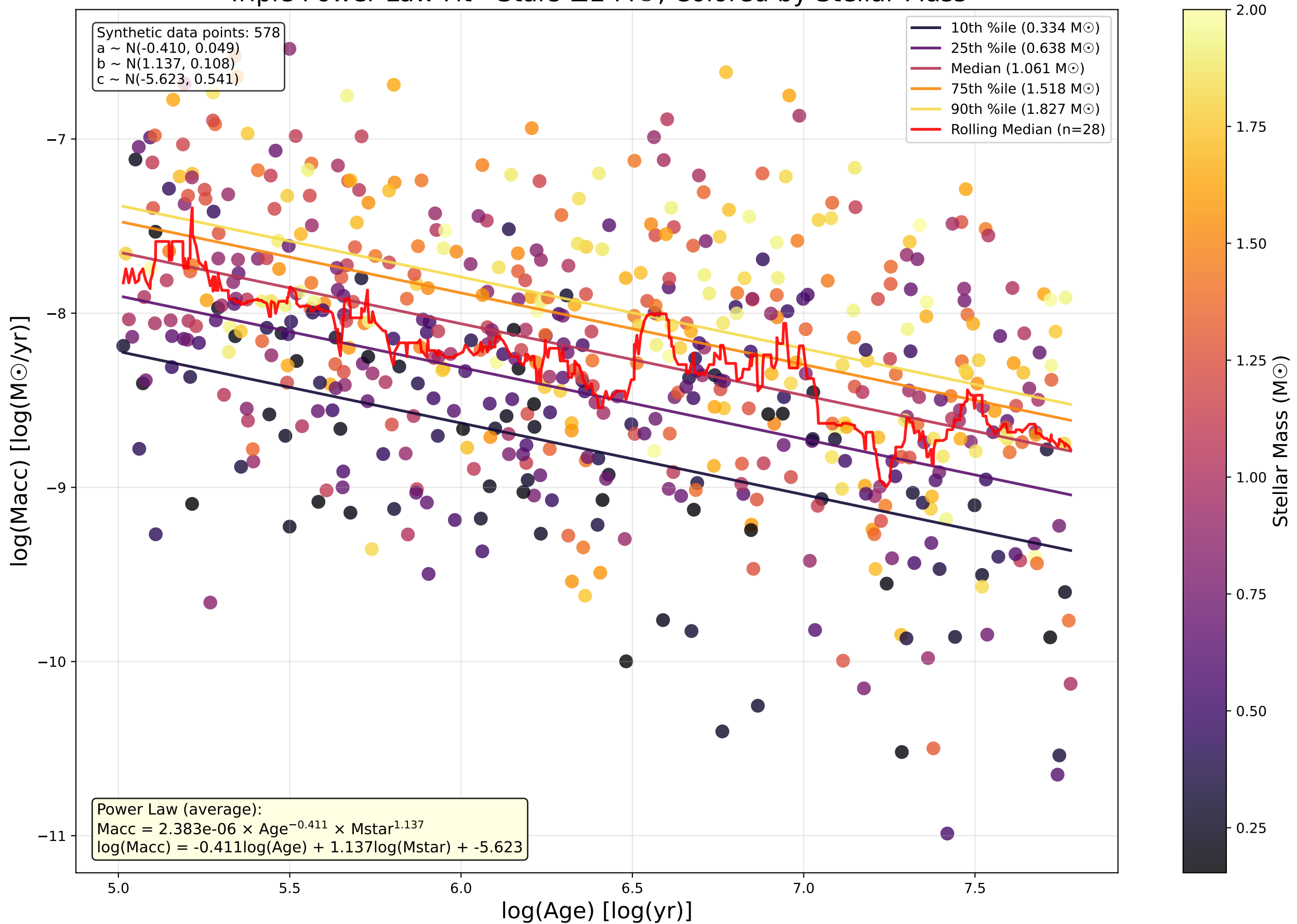


c-value Distribution



Synthetic Data with Normal Distribution Parameters

Triple Power Law Fit - Stars $\leq 2 M_{\odot}$, Colored by Stellar Mass



SYNTHETIC DATA WITH NORMAL DISTRIBUTION PARAMETERS

Complete Analysis Report

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METHOD:

- Each synthetic star gets its own (a,b,c) parameters
- Parameters drawn from normal distributions:
 - a ~ $N(\mu=-0.410, \sigma=0.049)$
 - b ~ $N(\mu=1.137, \sigma=0.108)$
 - c ~ $N(\mu=-5.623, \sigma=0.541)$
- $\log\text{Macc} = a \times \log\text{Age} + b \times \log\text{Mstar} + c$

STATISTICAL COMPARISON (KS tests):

logAge: p=0.000 x
Mstar: p=0.000 x
logMacc: p=0.000 x
c-values: p=0.554 ✓

KEY INSIGHTS:

1. Synthetic data captures statistical properties of real data
2. Parameter uncertainty adds realistic scatter to predictions
3. Normal distribution assumption validated by bootstrap
4. Complete uncertainty propagation model achieved